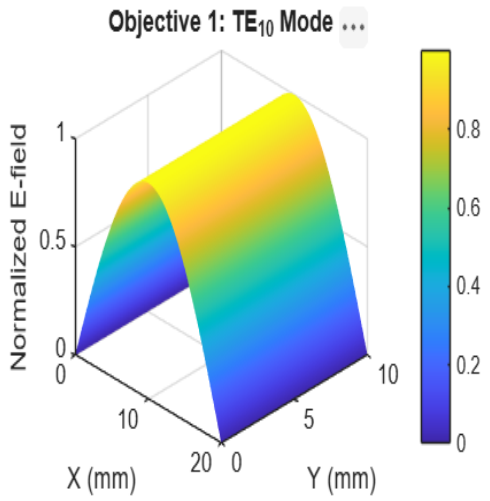
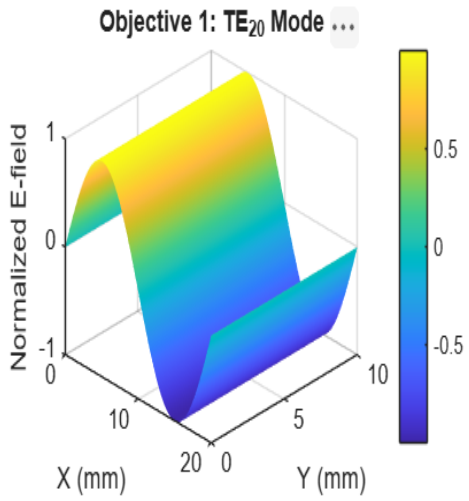
EXPERIMENT 1

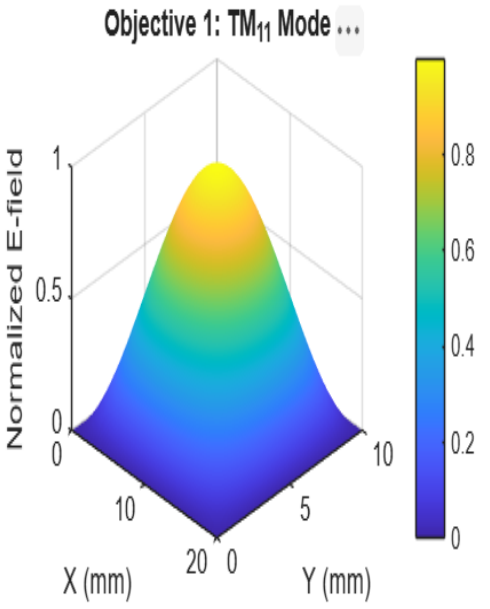
AIM : Simulation-Based Electromagnetic Analysis of TEM,TE and TM Propagation Modes in Microwave Transmission Structures

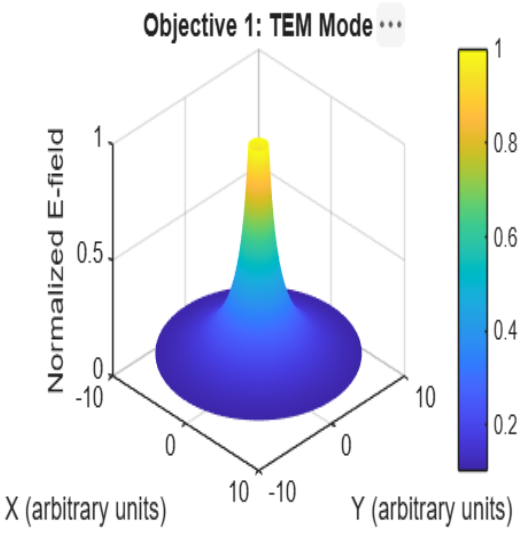
Objective 1 : To develop electromagnetic simulation models for TEM, TE and TM propagation modes in suitable microwave transmission structures.

Output:





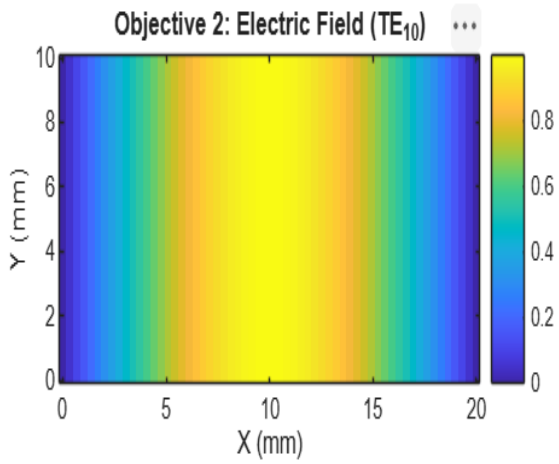


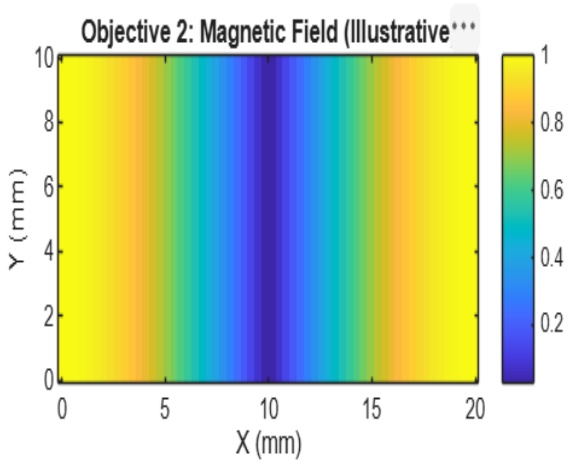


Objective 2:

To visualize and analyze the electric-field and magnetic-field distributions associated with different propagation modes.

Output:

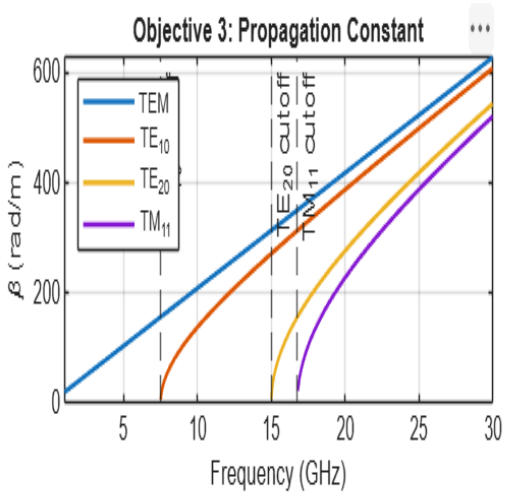


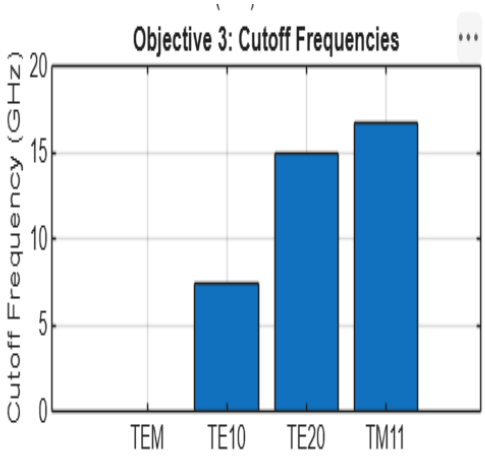


Objective 3:

To determine the cutoff frequency, propagation constant and field characteristics of selected TE and TM modes.

Output:

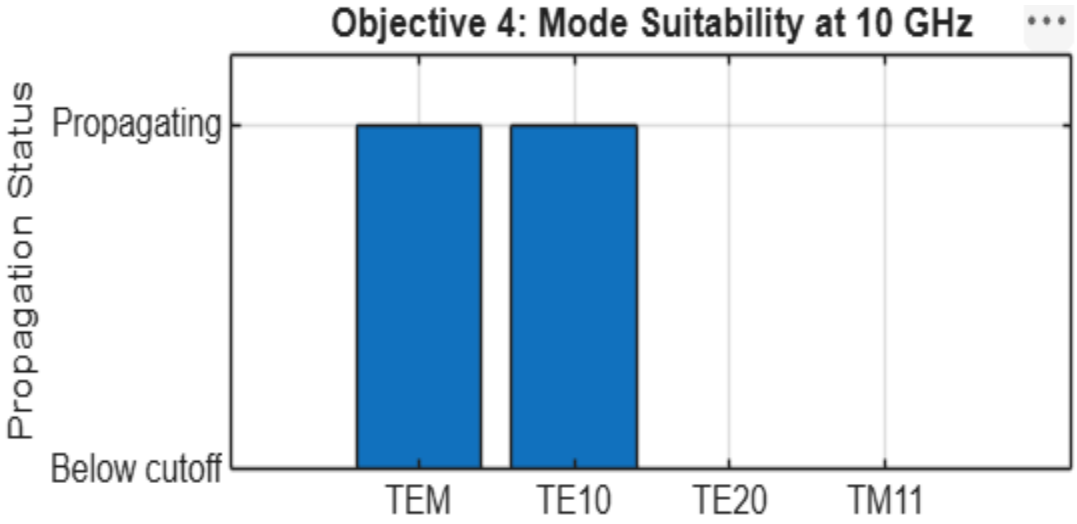




Objective 4:

To compare the propagation behavior of different modes and identify their suitability for practical microwave transmission.

Output:




```
========================================
 MICROWAVE ENGINEERING - EXPERIMENT 1
========================================
Operating Frequency = 10.00 GHz

CUTOFF FREQUENCIES
TEM  = 0 GHz (ideal two-conductor line)
TE10 = 7.500 GHz
TE20 = 15.000 GHz
TM11 = 16.771 GHz

PROPAGATION CONSTANTS AT 10.00 GHz
TEM  beta = 209.440 rad/m
TE10 beta = 138.531 rad/m
TE20 is below cutoff at 10.00 GHz
TM11 is below cutoff at 10.00 GHz

PROPAGATION STATUS AT 10.00 GHz
TE10: Propagating
TE20: Below cutoff
TM11: Below cutoff
```

RESULT:

The electromagnetic propagation modes TEM, TE_{10}, TE_{20}, and TM_{11} were simulated and compared using MATLAB. The electric-field and magnetic-field distributions were visualized, and the cutoff frequencies and propagation constants were analyzed.

The calculated cutoff frequencies for the given waveguide dimensions ($a = 0.02\,m$, $b = 0.01\,m$) are:

- TE_{10}: 7.5 GHz
- TE_{20}: 15 GHz
- TM_{11}: 16.77 GHz
- TEM: No cutoff frequency in an ideal two-conductor transmission line.

At the operating frequency of 10 GHz, the TE_{10} mode propagates, whereas TE_{20} and TM_{11} are below cutoff. The simulation demonstrates the differences in field distributions and propagation characteristics of the electromagnetic modes.